

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 1

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Case Study	
System Name	Koperasi KADA Online Platform System
Group Name	Tech EcoPioneer

SECTION A (SYSTEM'S OVERVIEW)		
1.1	Overall Description	
		<i><< Describe the system's overview. >></i>
1.2	Project Weaknesses or Improvement	
		<i><< List the project's weaknesses or improvement that can be made to the proposed system. Provide an explanation of the proposed module to overcome the weaknesses. >></i>
1.3	Proposed Module	
		<i><< Describe a new module to overcome the project's weaknesses. >></i>
SECTION B (DATABASE PLANNING AND DESIGN)		
2.1	Conceptual ERD	
		<i><< Draw a conceptual ERD to represent the proposed module. >></i>
2.2	Logical ERD	
		<i><<Provide logical ERD of your module in the proposed system.>></i>
2.3	Data Dictionary for Logical ERD	
		<i><< Provide the data dictionary based on the logical ERD in Subsection 2.2.>></i>
2.4	Relational Database Schema	
		<i><<Provide all relations database schema of your proposed module.>></i>
2.5	Normalization	
		<i><< Illustrate the normalization process up until BCNF to every relation listed in 2.5. At the end of this section, list all normalized relations achieved. >></i>

SECTION A

1. Overall Description

Our client is Lembaga Kemajuan Pertanian Kemubu Kelantan, more usually called KADA, representing the cooperative members who need an efficient and modern financial management system. The system proposed by this topic is called the Koperasi KADA Online Platform System, which is a web-based application designed to replace traditional manual processes with an intuitive, responsive, and digital e-system. The major objective of the system is to enhance the recording, maintaining, and managing of data involved in cooperative operations such as members' savings, loans, transactions, and financial reporting. It will help alleviate administrative pressures by automating these processes and improve communications among members, administrators, and stakeholders of the cooperatives.

Our aim is to build a system that will meet the client's needs while incorporating efficiency and reliability. The system proposed will integrate five main modules, which are designed to address issues in cooperative management. These modules ensure the system operates completely functional, with efficiency and effectiveness, meeting the changing needs of the membership and administrators involved. Below, we provide a detailed overview of the modules and their functionalities.

The Login Module is the first module that provides secure and user-friendly authentication for members and administrators. This module grants access to the system to only registered users through OTP-based authentication. The Member Management Module acts as the backbone of the system, whereby an administrator can add, update, and manage member information, including personal details and membership status. This module makes sure that the members' data is correct and up-to-date, hence ensuring smooth operations and increased member engagement. The Member Savings Module enables members to manage their savings accounts with ease: creating recurring payments, viewing balances, and viewing detailed transaction histories in promoting financial transparency and convenience.

The Loan Module is another important component of the system. It also simplifies loan applications by offering submission, approval, and tracking tools. The module also contains an inbuilt loan calculator that would aid members in better financial planning and management. Reports Module: Offers extensive insight into the financial activities and gives detailed reporting capabilities to the administrators and program organizers for better decision-making and accountability. The purpose of this module is to let users observe and estimate the financial activities of the cooperative in a clean, well-organized presentation format.

This system features extensibility and maintainability: automated validation of documents; control of different user roles for access rights; and supporting access, allowing the software platform to adjust in the light of future requirements. The system is designed to easily modify and upgrade by abstracting the key data models and processes, which will enable the system to accommodate the client's long-term planning needs. The Koperasi KADA Online Platform System is a modern transformation in cooperative management, with its strong architecture and user-oriented design.

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NG YU HIN	A23CS0148	LOANS MANAGEMENT
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2. Project Weakness or Improvement

While the Koperasi KADA Online Platform System is a comprehensive and efficient solution, it has certain critical weaknesses that must be addressed to ensure optimal performance and reliability.

One significant weakness is the **absence of a robust mechanism** for real-time error detection and tracking. Without this capability, issues such as failed transactions, synchronization errors, or connectivity disruptions may go unnoticed until they escalate into larger problems. This can lead to delays in resolution, prolonged system downtimes, and overall inefficiency in operations. To improve this, the system should incorporate automated error detection tools that can promptly identify and flag system issues for timely resolution.

One major limitation is that the **system lacks central error logging**. Currently, the system does not have a central place to record system errors. Instead, the current system will force administrators to depend on ad hoc and manual means to track down problems and fix them. This not only increases the time and effort required to resolve issues but also makes it challenging to analyze patterns or recurring problems that could inform long-term solutions. Establishing a centralized error logging system would streamline error management and provide valuable insights into system performance over time.

The system also suffers from an **inadequate notification mechanism** to alert administrators about critical errors or issues. This means that problems are often discovered too late, when they have already affected the system's functionality or user experience. Without timely alerts, administrators cannot proactively address problems, increasing the risk of disruptions and reducing user trust in the system's reliability. Introducing real-time notification features would ensure that administrators are promptly informed about critical issues, enabling faster response times and minimizing disruptions.

Additionally, the absence of **historical reporting on system errors** hinders the ability to implement preventive measures. Without access to detailed records of past errors, it is difficult for administrators to identify recurring problems or trends that could guide improvements to system reliability. Incorporating historical reporting functionality would

allow administrators to analyze recurring issues, understand root causes, and implement preventive measures to enhance system stability.

Lastly, the system lacks a structured approach to **tracking the resolution of errors**. Without a mechanism to monitor the progress of issue resolution, administrators may struggle to ensure accountability and transparency in the error management process. This can lead to delays in resolving issues and a lack of clarity about the steps being taken to address them. Implementing a resolution tracking system would ensure efficient problem-solving and maintain transparency for all stakeholders.

By addressing these weaknesses by targeted improvements, the system could achieve greater reliability, operational efficiency, and user satisfaction. When measures to effectively detect, log, notify, and track errors are implemented, it will help minimize disruptions to the system as well as further enhance the users' trust and confidence in using the platform.

3. Proposed Module

The **Error Monitoring Module** is a vital addition aimed at enhancing the system's reliability and efficiency of operation. It identifies direct weaknesses in error identification and resolution, adding a simplified procedure for tracking, monitoring, and managing system errors. This module will make sure that issues are reported to system administrators on time, thus reducing downtimes and keeping the integrity of the system.

At the core of the functionality of this module is **real-time error detection**. It identifies a failure in data synchronization, some transactional error, or the loss of connectivity. This logs errors automatically through an extensive **error logging system**, which captures time and date, error type, description, and system component in question. This forms a trustworthy repository for the diagnosis of a problem and identification of trends in recurring issues.

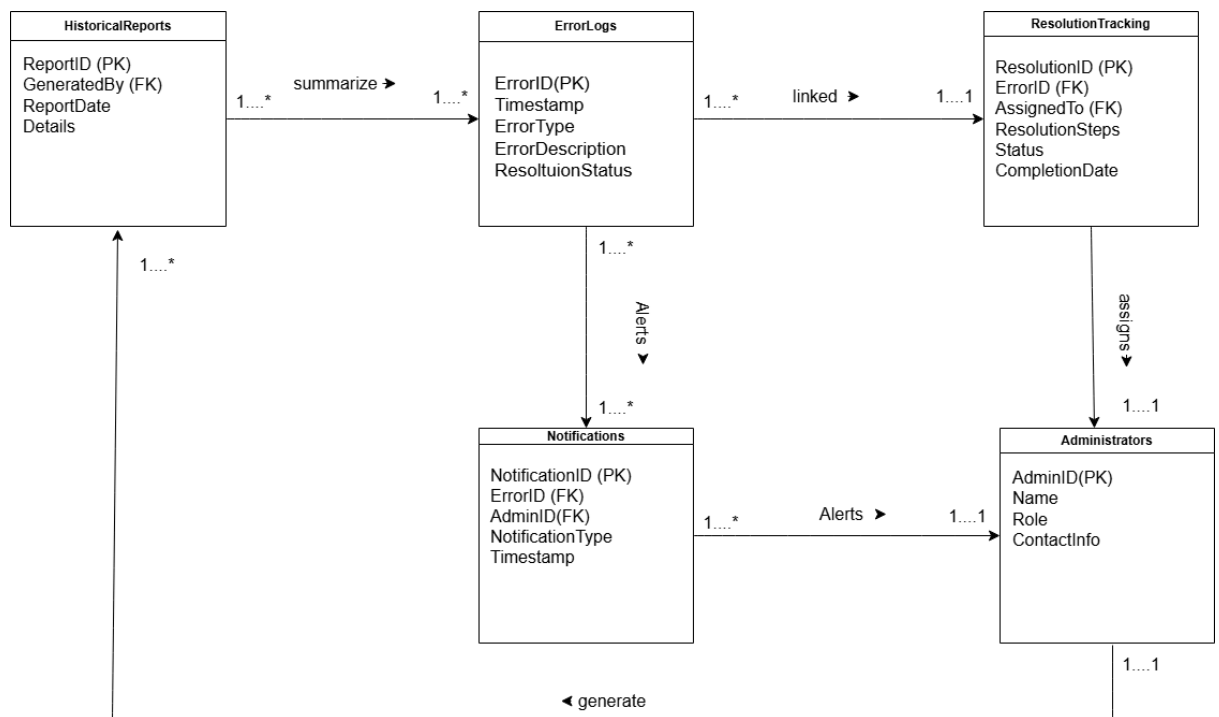
The module features a **live notification system** designed to alert an administrator in real time when some critical issues have been detected. Notifications can then be relayed through email, SMS, or in-system, ensuring that whichever channel is considered best, notifications will always go through to ensure the administrator is immediately informed, whatever their location. This reduces response times and speeds up error resolution processes. Moreover, the module supports **resolution status tracking**, enabling the administrator to see the progress of fixing an issue. This ensures accountability and gives transparency to the steps being taken in order to solve the problem.

To further improve system reliability, the module includes a **historical error reporting feature**. This feature generates detailed reports on past errors, including their frequency, resolution times, and associated impacts. These insights help administrators identify patterns and root causes, enabling proactive measures to prevent future occurrences. The module is designed to be user-friendly, with a dashboard that provides a clear overview of all logged errors, their statuses, and relevant trends.

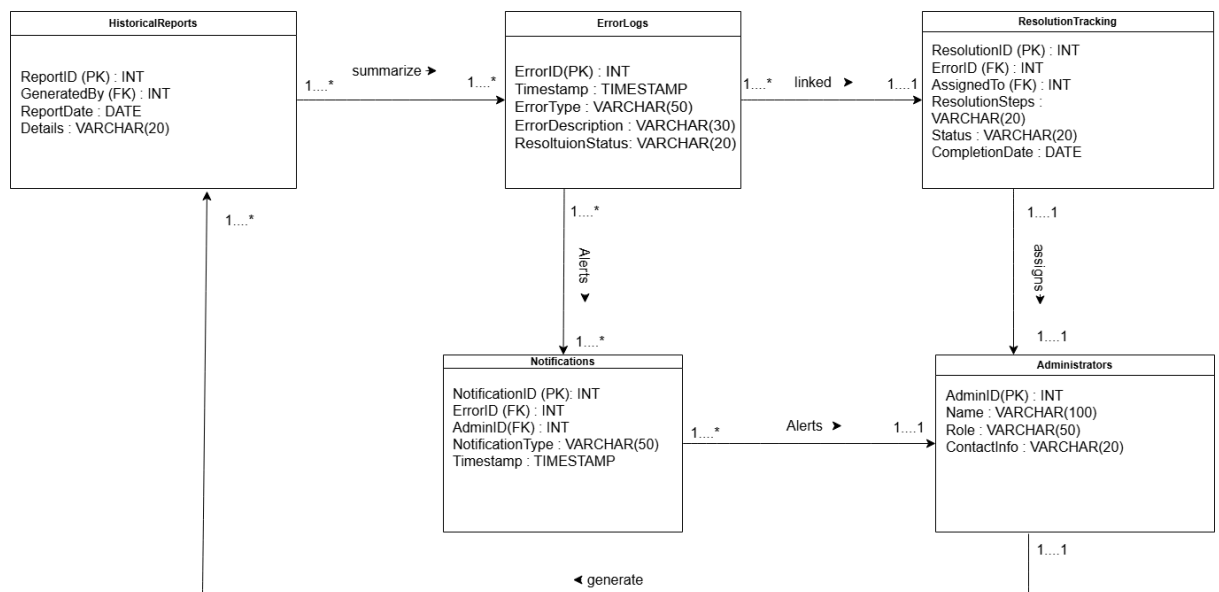
The **Error Monitoring Module** will bring the system close to operational reliability; possible periods of downtime will be significantly reduced. The module will not only minimize disruptions but also instill greater confidence in the system among users and administrators. Its robust functionality and proactive design make it a valuable addition to any digital platform requiring enhanced performance and error management.

SECTION B

4. CONCEPTUAL ERD



5. LOGICAL ERD



6. DATA DICTIONARY FOR LOGICAL ERD

Table Name	Field Name	Data Type	Description
ErrorLogs	ErrorID (PK)	INT	Unique identifier for each error
	Timestamp	TIMESTAMP	The date and time the error occurred
	ErrorType	VARCHAR(50)	The type of error (e.g., Sync Error)
	ErrorDescription	VARCHAR(30)	Detailed description of the error
	ResolutionStatus	VARCHAR(20)	Status of the error (e.g., Resolved, Pending)
Administrators	AdminID (PK)	INT	Unique identifier for each administrator
	Name	VARCHAR(50)	Name of the administrator
	Role	VARCHAR(50)	Role in the system (e.g., System Admin)
	ContactInfo	VARCHAR(20)	Contact details of the administrator
Notifications	NotificationID (PK)	INT	Unique identifier for notifications
	ErrorID (FK)	INT	Associated error ID
	AdminID (FK)	INT	Admin receiving the notification
	NotificationType	VARCHAR(50)	Type of notification (e.g., Email, SMS)
	Timestamp	TIMESTAMP	The time the notification was sent
HistoricalReports	ReportID (PK)	INT	Unique identifier for each report
	GeneratedBy (FK)	INT	Admin who generated the report
	ReportDate	DATE	Date of report generation

	Details	VARCHAR(20)	Content and summary of the report
ResolutionTracking	ResolutionID (PK)	INT	Unique identifier for each resolution
	ErrorID (FK)	INT	Error being resolved
	AssignedTo (FK)	INT	Admin responsible for resolving the error
	ResolutionSteps	VARCHAR(20)	Steps taken to resolve the error
	Status	VARCHAR(20)	Current resolution status
	CompletionDate	DATE	Date when the resolution was completed

7. RELATIONAL DATABASE SCHEMA

Relation Name	Attributes	Primary Key (PK)	Foreign Key (FK)
ErrorLogs	ErrorID, Timestamp, ErrorType, ErrorDescription, ResolutionStatus	ErrorID	-
Administrators	AdminID, Name, Role, ContactInfo	AdminID	-
Notifications	NotificationID, ErrorID, AdminID, NotificationType, Timestamp	NotificationID	ErrorID → ErrorLogs.ErrorID AdminID → Administrators.AdminID
HistoricalReports	ReportID, GeneratedBy, ReportDate, Details	ReportID	GeneratedBy → Administrators.AdminID
ResolutionTracking	ResolutionID, ErrorID, AssignedTo, ResolutionSteps, Status, CompletionDate	ResolutionID	ErrorID → ErrorLogs.ErrorID AssignedTo → Administrators.AdminID

8. Normalization

Relation	Step	Explanation	Resulting Relation
ErrorLogs	UNF	Potentially contains non-atomic values or repeating groups.	ErrorID, Timestamp, ErrorType, ErrorDescription, ResolutionStatus
	1NF	Ensure all attributes are atomic and no repeating groups exist.	ErrorID (PK), Timestamp, ErrorType, ErrorDescription, ResolutionStatus
	2NF	All non-key attributes are fully functionally dependent on the primary key.	Already in 2NF since all attributes depend on ErrorID
	3NF	Ensure no transitive dependency exists.	Already in 3NF since there are no transitive dependencies.
	BCNF	Ensure every determinant is a candidate key.	Already in BCNF as ErrorID is the only determinant.
Administrators	UNF	Potentially contains non-atomic values or repeating groups.	AdminID, Name, Role, ContactInfo
	1NF	Ensure all attributes are atomic and no repeating groups exist.	AdminID (PK), Name, Role, ContactInfo
	2NF	All non-key attributes are fully functionally dependent on the primary key.	Already in 2NF since all attributes depend on AdminID
	3NF	Ensure no transitive dependency exists.	Already in 3NF since there are no transitive dependencies.

	BCNF	Ensure every determinant is a candidate key.	Already in BCNF as AdminID is the only determinant
Notifications	UNF	Potentially contains non-atomic values or repeating groups.	NotificationID, ErrorID, AdminID, NotificationType, Timestamp
	1NF	Ensure all attributes are atomic and no repeating groups exist	NotificationID (PK), ErrorID (FK), AdminID (FK), NotificationType, Timestamp
	2NF	All non-key attributes are fully functionally dependent on the primary key.	Already in 2NF since all attributes depend on NotificationID.
	3NF	Ensure no transitive dependency exists	Already in 3NF since there are no transitive dependencies.
	BCNF	Ensure every determinant is a candidate key.	Already in BCNF as NotificationID is the only determinant.
HistoricalReports	UNF	Potentially contains non-atomic values or repeating groups.	ReportID, GeneratedBy, ReportDate, Details
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ResolutionTracking	UNF	Potentially contains non-atomic values or repeating groups.	ResolutionID, ErrorID, AssignedTo, ResolutionSteps, Status, CompletionDate
	1NF	Ensure all attributes are atomic and no repeating groups exist.	ResolutionID (PK), ErrorID (FK), AssignedTo (FK), ResolutionSteps, Status, CompletionDate
	2NF	All non-key attributes are fully functionally dependent on the primary key.	Already in 2NF since all attributes depend on ResolutionID.
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